

Clifford algebras & Spin groups

Let $Cl(k) = Cl(\mathbb{R}^k) = \mathbb{R}^k / \langle v \otimes v + (v,v)1 : v \in \mathbb{R}^k \rangle$. Note that $Cl(k)^\times = Cl(k) \setminus 0$.

Define $Cl(k)_0 = \langle v \otimes w : v, w \in \mathbb{R}^k \rangle_{\text{mult.}}$, the even subalgebra. Then

$Pin(k) = \langle S^{k-1} \rangle_{\text{mult.}}$, the group generated by the unit vectors,
 $Spin(k) = Pin(k) \cap Cl(k)_0$.

Note that for $v \in S^{k-1}$, $v^{-1} = -v$ in $Cl(k)$. We define a representation

$$\rho: Pin(k) \rightarrow Aut(Cl(k)), \quad \rho(y): x \mapsto yx\varepsilon(y^{-1})$$

on the generators $y \in S^{k-1} \in Pin(k)$, where $\varepsilon: Cl(k) \rightarrow Cl(k)$ is the grading automorphism, which acts as 1 on $Cl(k)_0$ and as -1 on $Cl(k)_1$. Hence,

$$\rho(y): x \mapsto -yx y^{-1} = yxy$$

on S^{k-1} . Note that for $y \in S^{k-1}$, $x \in \mathbb{R}^k$

$$\begin{aligned} \rho(y)x &= yx\varepsilon(y^{-1}) = yxy = y((x_1 y) + \overbrace{(x - (x_1 y))}^v) y \\ &= y(x_1 y) y^2 + yvy \\ &= -(x_1 y) y - vy^2 \quad (y^2 = -1, vy = -\overbrace{(y,v)}^0 - vy) \\ &= -(x_1 y) y + v \\ &= x - 2(x_1 y) y. \end{aligned}$$

Hence, $\rho(y)|_{\mathbb{R}^k} \in O(k)$ is a reflection.

Prop: We have a s.e.s.

$$0 \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow Spin(k) \xrightarrow{\rho} SO(k) \rightarrow 0.$$

Proof:

• ρ is surjective: $SO(k)$ is generated by products of even numbers of reflections

• $\ker \rho = \mathbb{Z}/2\mathbb{Z} = \{\pm 1\}$:

$$\ker \rho = \{y \in Spin(k) : [y, v]_S = 0 \quad \forall v \in \mathbb{R}^k\} \subseteq \{x \in Cl(k) : xv = (-1)^{\deg x} vx = 0 \quad \forall v \in \mathbb{R}^k\}$$

Since $\mathbb{R}^k$ generates $Cl(k)$.

- $\{x \in Cl(k) : xv = (-1)^{\deg x} vx = 0 \quad \forall v \in \mathbb{R}^k\} = \{0\}$. Write $x = a + e_1 b$. Wlog, x is homogeneous.

Then $\deg x = \deg a = \deg b + 1$ and

$$\begin{aligned} x e_1 &= a e_1 + e_1 b e_1 \\ &= (-1)^{\deg x} (e_1 a - e_1 b) \\ &= (-1)^{\deg x} (e_1 a + b). \end{aligned}$$

Since $e_1 x = e_1 a - b$, we need $b = 0$, so x does not involve e_1 , etc.

- $Spin(k) \cap \mathbb{R} = \{\pm 1\}$: We have an antiautomorphism $Cl(k) \xrightarrow{t} Cl(k)$ extending

$$v_1 \dots v_m \mapsto v_m \dots v_1$$

on $\mathbb{R}^k = \text{Sp}(v_1, \dots, v_m)$. On $Spin(k)$, this is inversion, so on $Spin(k) \cap \mathbb{R}^k$, we

get $x^{-1} = x^t = x$. $\square$

Corollary: For $k \geq 2$, $\pi_1(Spin(k)) = 1$.

Proof: We have

$$SO(n) \hookrightarrow SO(n+1)$$

$$\downarrow$$

$$S^1$$

so $\pi_2(S^1) \rightarrow \pi_1(SO(n)) \rightarrow \pi_1(SO(n+1)) \rightarrow \pi_1(S^1)$ is exact. Hence, for $n \geq 2$,

$$\pi_1(SO(n)) = \pi_1(SO(n+1)).$$

But $SO(3) \cong \mathbb{R}P^3$, by taking a rotation to a vector in $\mathbb{R}^3$ with (length, direction) = (angle, oriented axis).

Hence, $\pi_1(SO(3)) = \mathbb{Z}/2\mathbb{Z}$. $\square$

Representations of $Cl(k) \otimes \mathbb{C}$

we will always mean this when we talk about representations of $Cl(k)$

Let $E_k \subseteq Pin(k)$ be the subgroup (of order 2^{k+1}) consisting of

$$\pm e_1^{i_1} \dots e_k^{i_k},$$

where $\{e_1, \dots, e_k\} \subset \mathbb{R}^k$ is an orthonormal basis, and $i_j \in \{0, 1\}$. Let $v = -1 \in E_k$. Then

$$\begin{aligned} \{\text{algebra reps of } Cl(k)\} &= \{\text{group reps of } Pin(k) \text{ s.t. } v \text{ acts as } -1\} \\ &= \{\text{group reps of } E_k \text{ s.t. } v \text{ acts as } -1\}. \end{aligned}$$

Hence, we can study the representations of E_k . Note that v acts as +1 or -1 on any irreducible representation (since it is central and an involution). We get 2^k representations of the abelian group E_k/v .

We have $\# \text{ irreps} = \# \text{ conj. classes}$. Since E_k/v is abelian, the conjugacy classes are $\{g\}$ (g central), $\{g, vg\}$ (g not central).

Lemma: $Z(E_{2m}) = \{1, v\}$

$$Z(E_{2m+1}) = \{1, v, w, vw\} \quad w = e_1 \dots e_m$$

Hence, the number of conjugacy classes are

$$\frac{2^{k+1} - 2}{2} + 2 = 2^k + 1 \quad \text{and} \quad \frac{2^{k+1} - 4}{2} + 4 = 2^k + 2$$

k even

k odd.

So in the even case, E_k only has one irreducible representation on which $v = -1$. We denote it by Δ and call it the spin representation of $Cl(k)$. Since the sum of the squares of the dimensions of the irreps of E_k is $|E_k|$, we get

$$2^k + (\dim \Delta)^2 = 2^{k+1},$$

ie $\dim \Delta = 2^{m/2}$.

Any finite dimensional representation W of $Cl(k)$ is of the form $\Delta \otimes V$ for some vector space V . Note that

$$V = \text{Hom}_{Cl(k)}(\Delta, W) \quad \text{and} \quad \text{End}_{Cl(k) \otimes \mathbb{C}} W = \text{End}_{\mathbb{C}} V.$$

If $F \in \text{End}_{Cl(k) \otimes \mathbb{C}} W$, we define its relative trace $\text{tr}^{W/\Delta} F$ to be the trace of the corresponding endomorphism of V .

Note: Since Δ is a representation of $Pin(k)$, it gives a representation of $Spin(k)$.

As such, it splits into a sum $\Delta_+ \oplus \Delta_-$ of irreducible representations, the ± 1 -spaces for $w = e_1 \dots e_m$. Note that $\Delta_+ \oplus \Delta_-$ is also a supergrading for $Cl(k) \otimes \Delta$.

Clifford bundles & Dirac operators

The discussion can be done globally for the bundle TM on a Riemannian manifold.
 ie $Cl(TM) \otimes \mathbb{C} \otimes S$

Def: Let S be a bundle of Clifford modules on M . Then S is a **Clifford bundle** if

- S has a Hermitian metric (and compatible connection ∇^S);
- for $v \in TM$, $s_1, s_2 \in S_v$
 $(v \cdot s_1, s_2) + (s_1, v \cdot s_2) = 0$;
- for $X, Y \in C^\infty(TM)$, $s \in C^\infty(S)$,
 $\nabla_X^S (Ys) = (\nabla_X^{TM} Y)s + Y \cdot \nabla_X^S s$.

We get a Dirac operator D as the composition

$$C^\infty(S) \xrightarrow{\nabla^S} C^\infty(T^*M \otimes S) \xrightarrow{(\cdot, \cdot)_{TM}} C^\infty(TM \otimes S) \xrightarrow{Cl(TM) \otimes S} C^\infty(S).$$

In coordinates,

$$s \mapsto \nabla^S s = \sum e_i \otimes \nabla_{e_i} s \mapsto \sum e_i \otimes \nabla_{e_i} s \mapsto \sum e_i \nabla_{e_i} s.$$

($e_i, \cdot$)
orthonormal basis

Note that for e_i orthonormal s.t. $[e_i, e_j] = \nabla_{e_i} e_j = 0$

$$D^2 s = \sum e_j \nabla_j (e_i \nabla_i s)$$

$$= \sum (e_j \nabla_j e_i \nabla_i s + e_j e_i \nabla_j \nabla_i s)$$

$$= - \sum \nabla_i^2 s + \sum_{j < i} e_j e_i (\underbrace{\nabla_j \nabla_i - \nabla_i \nabla_j}_{K(e_j, e_i)}) s \quad e_i e_j + e_j e_i = -2\delta_{ij}$$

This needs to be checked $\rightarrow$

$$= \nabla^* \nabla s + Ks. \quad K = \sum_{j < i} e_j e_i K(e_j, e_i) \in C^\infty(S)$$

One can check that for $s_1, s_2 \in C^\infty(S)$, $(Ds_1, s_2) = (s_1, Ds_2)$.

Let us analyse K further. Let $K(X, Y) = \nabla_X^S \nabla_Y^S - \nabla_Y^S \nabla_X^S - \nabla_{[X, Y]}^S \in C^\infty(S)$ be the Clifford curvature, R the Riemannian curvature. One can check that

$$[K(X, Y), c(Z)] = c(R(X, Y)Z),$$

where $c: Cl(TM) \rightarrow \text{End } C^\infty(S)$ is the Clifford action. The same is true if we replace $K(X, Y)$ by $R^S(X, Y) = \frac{1}{4} \sum c(e_k) c(e_l) (R(X, Y) e_k e_l)$,

where $\{e_i\}$ is an orthonormal basis. Hence,

$$K = R^S + F^S,$$

where F^S , the **twisting curvature**, commutes with the Clifford action. One can check that

$$D^2 = \nabla^* \nabla + F^S + \frac{1}{4} K \quad \leftarrow \text{scalar curvature}$$

$\uparrow \sum_{i < j} c(e_i) c(e_j) F^S(e_i, e_j)$

Examples of Clifford modules:

First, we recall the Hodge star $*$: $\Lambda^k T^*M \rightarrow \Lambda^{n-k} T^*M$, defined so that

$$\beta \wedge * \alpha = (\alpha, \beta) \text{ vol.}$$

Then $** = (-1)^{k(n-k)}$. Also, $\flat^*: \Lambda^k T^*M \rightarrow \Lambda^{k-1} T^*M$ is given by $\flat^* = \pm * \flat$. Note that $* \text{vol} = 1$. For $e \in C^\infty(T^*M)$, $w \in C^\infty(\Lambda^k T^*M)$, define

$$e \lrcorner w = (-1)^{k(n-k)+1} * (e \wedge * w) \in C^\infty(\Lambda^{k-1} T^*M).$$

Note that if $p, q \in C^\infty(T^*M)$ then $q \lrcorner p = -(p, q)$.

Example 1 ($Cl(TM) \otimes \Lambda^* T^*M \otimes \mathbb{C}$): This action is the usual left multiplication of $Cl(TM) = Cl(T^*M)$ on itself together with the vector space isomorphism

$$Cl(T^*M) \rightarrow \Lambda^* T^*M.$$

It is given for $e \in C^\infty(T^*M)$, $w \in \Lambda^* T^*M$ by $e \cdot w = e \wedge w + e \lrcorner w$. Check that this is really a Clifford bundle. Its Dirac operator is

$$Dw = \sum c(e_i) \nabla_i w = \sum e_i \wedge \nabla_i w + e_i \lrcorner \nabla_i w = \flat w + \flat^* w. \quad \leftarrow \text{use Cartan's formula}$$

Example 2 (Complex manifolds): Assume TM has a complex structure J compatible with the metric. Then

$$T^*M \otimes \mathbb{C} = P \oplus Q,$$

where P, Q are the $\pm i$ -spaces for J . Note that they are isotropic.

Let $p \in C^\infty(P)$, $q \in C^\infty(Q)$, $x \in C^\infty(\Lambda^* P)$. Define

$$(p + q) \cdot x = \sqrt{2} (p \wedge x + q \lrcorner x).$$

Since $p^2 = q^2 = 0$ and

$$\begin{aligned} (pq + qp) \cdot x &= pq \cdot x + qp \cdot x \\ &= 2p \wedge (q \lrcorner x) + 2q \lrcorner (p \wedge x) \\ &= 2(p \wedge (q \lrcorner x) + (q \lrcorner p) \wedge x - p \wedge (q \lrcorner x)) \\ &= 2(q \lrcorner p) \wedge x = -2(p, q)x, \end{aligned}$$

This extends to an action $Cl(TM) \otimes \Lambda^* P$. Note that $\dim \Lambda^* P = 2^k$, so this is fibre-wise the spin representation Δ !

Note that $S = \Lambda^* P = \Lambda^* T_{\mathbb{C}} M = \Lambda^* \overline{T^* M}$ is the bundle of antiholomorphic forms, ie

$$C^\infty(S) = \bigoplus \Omega^{0, q}(M).$$

One can check that $D = \sqrt{2} (\bar{\partial} + \bar{\partial}^*)$ on a Kähler manifold.

Spin structures

If M is ^{an oriented} Riemannian manifold, it has an $SO(n)$ -frame bundle E . A spin-structure on M is a principal $Spin(n)$ -bundle $\tilde{E}$ which is fibre wise the double covering $Spin(n) \rightarrow SO(n)$

Prop: M has a spin structure iff its second Stiefel-Whitney class $w_2(M) = 0 \in H^2(M; \mathbb{Z}/2\mathbb{Z})$.

Method 1: From the s.e.s. $0 \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow Spin(n) \rightarrow SO(n) \rightarrow 0$, we get an exact sequence

$$H^1(M; \mathbb{Z}/2\mathbb{Z}) \rightarrow H^1(M; Spin(n)) \rightarrow H^1(M; SO(n)) \rightarrow H^2(M; \mathbb{Z}/2\mathbb{Z}).$$

Method 2: We want

$$\begin{array}{ccc} & BSpin(n) & \\ \swarrow & \downarrow & \searrow \\ M & \rightarrow & BSO(n). \end{array}$$

fibres $B\mathbb{Z}_2 = \mathbb{RP}^1$

The pullback of $w_2 \in H^2(BSO(n))$ in $H^2(BSpin(n))$ is 0.

Method 3: The E_2 -page of the Serre spectral sequence associated to $SO(n) \hookrightarrow E$

$$\begin{array}{ccc} & & E \\ & & \downarrow \\ & & M \end{array}$$

take coefficients in $\mathbb{Z}/2\mathbb{Z}$

$H^1(SO(n))$ glue by an action of $\text{Maps}(U, SO(n))$ - take U trivial to get $\text{Maps}(pt, SO(n))$

has $E_2^{pq} = H^q(M; H^p(SO(n))) = H^q(M) \otimes H^p(SO(n))$. Hence,

$$\begin{array}{ccc} E_2^{20} & & \\ E_2^{40} & \xrightarrow{\quad} & E_2^{41} \\ E_2^{\infty} & \xrightarrow{\quad} & E_2^{\infty 0} \end{array} = \begin{array}{ccc} H^1(SO(n)) & & \\ \mathbb{Z}/2\mathbb{Z} & H^1(M) & H^2(M) \end{array}$$

Since $E_2^{pq} \Rightarrow H^{p+q}(E)$, we get

- $E_2^{01} \hookrightarrow H^1(E)$ (E_2^{01} is already degenerate)
- $\text{Coker } \pi^* = \text{Ker } (E_2^{40} \rightarrow E_2^{02})$. (ie E_2^{40} is degenerate)

Hence, we get an exact sequence (coefficients in $\mathbb{Z}/2\mathbb{Z}$)

$$0 \rightarrow H^1(M) \xrightarrow{\pi^*} H^1(E) \xrightarrow{i^*} H^1(SO(n)) \xrightarrow{\delta} H^2(M).$$

$H^1(E) = E_2^{40} \oplus E_2^{01}$ $\text{Ker } \delta = E_2^{40}$

Now $H^1(E; \mathbb{Z}/2\mathbb{Z})$ is the set of double coverings of E . Spin structures are $\text{Im } i^* = \text{Ker } \delta$.

We get an associated bundle with fibres Δ , which we also denote by Δ . It comes with a Hermitian metric (since $Spin(n) \rightarrow U(\Delta)$) and the pullback connection from $\tilde{E} \rightarrow E$. Also, $\text{Cl}(TM) \otimes \Delta$.

Prop: Δ is a Clifford bundle with twisting curvature $F^\Delta = 0$.

Note: If S is any Clifford bundle on M then $S = \Delta \otimes V$ for $V = \text{Hom}_{\text{Cl}}(\Delta, S)$ and its curvature is

$$K^\Delta \otimes 1 + 1 \otimes K^V.$$

Hence, its twisting curvature is K^V .

Characteristic classes of spin bundles

Here, let M be a spin manifold of dimension $2m$.

ie the Chern genus of $\mathbb{R}^{2m}$ of its complexification

Prop: The Chern character $ch(\Delta)$ is $2^m \text{G}(TM)$, where G is the Pontrjagin genus associated to $2 \mapsto \cosh(\frac{1}{2}\sqrt{2})$

Let S be a Clifford bundle on M . Then its relative Chern character is the class

$$ch(S/\Delta) = \text{tr}^{S/\Delta} \left(\exp(-\frac{1}{2\pi i} F^S) \right).$$

Note that if $S = \Delta \otimes V$ then $ch(S/\Delta) = ch(V)$. From the proposition, since

$$ch(\Delta \otimes V) = ch(\Delta) ch(V), \quad ch(S) = 2^m \text{G}(TM) ch(S/\Delta).$$

The complex Spin group

We let $\text{Spin}^c(k) = \langle \text{Spin}(k), \overset{\mathbb{C}}{S^1} \rangle_{\text{mult}} \subseteq \text{Cl}(k) \otimes \mathbb{C}$.

Since $S^1 \subseteq Z(\text{Cl}(k) \otimes \mathbb{C})$, $\text{Spin}(k) \times S^1 \rightarrow \text{Spin}^c(k)$ with kernel (λ^{-1}, λ) for $\lambda \in S^1 \cap \text{Spin}(k) = \{\pm 1\}$. Note that for even k , Δ is also a representation of $\text{Spin}^c(k)$.

Prop: We have a s.e.s.

$$0 \rightarrow \mathbb{Z}/2\mathbb{Z} \xrightarrow{1 \mapsto (-1, 1)} \text{Spin}^c(k) \xrightarrow{(x, \lambda) \mapsto (q(x), \lambda^2)} \text{SO}(k) \times S^1 \rightarrow 1.$$

$\swarrow q: \text{Spin}(k) \rightarrow \text{SO}(k)$

We call a $\text{Spin}^c(n)$ -bundle $\tilde{E}$ over M a Spin^c -structure with fundamental line bundle L if it is fibrewise the double covering $\text{Spin}^c(k) \rightarrow \text{SO}(k) \times S^1$. (Choose a connection on L and lift it to a connection on $\Delta(\tilde{E})$).

Note: If M is a Riemannian manifold with a Clifford bundle S whose fibres are the spin representations then it can be shown that it comes from a Spin^c -structure.